

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative analytics researcher with experience developing and validating stochastic and machine-learning models for financial risk and large dependent datasets. Builds Python/SQL/PyTorch workflows for simulation, optimization, model diagnostics, and performance evaluation on AWS and HPC.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL/MySQL, R, Linux, Git, AWS EC2, HPC/SLURM

Quantitative Analytics: Stochastic modeling, optimization, Monte Carlo simulation, computational statistics, predictive modeling, machine learning, model development and validation, performance assessment and evaluation testing

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Developed and validated a sequential anomaly-detection framework for binary risk indicators by representing cumulative evidence as a random walk and using Brownian-motion approximations to quantify stopping-time uncertainty.
- Performed simulation-based comparisons of fixed-sample and sequential rules across detection delay, false-positive/false-negative control, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM predictive-modeling pipeline for mid-market property risk, including data processing, GLM benchmarking, validation, and performance diagnostics across 2.46M+ records on AWS EC2.
- Translated model diagnostics and feature-attribution results into pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE MODEL DEVELOPMENT AND VALIDATION

Ph.D. Dissertation Research - Scalable Gaussian Process Inference and Diagnostics

Sep 2024 - Present

- **Model Development:** Developed an advection-aware Vecchia likelihood for nonseparable spatiotemporal Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- **Calibration and Validation:** Calibrated Matérn and generalized Cauchy covariance models and built scale- and frequency-resolved diagnostics to detect misspecification in smoothness, noise, and dependence.
- **Simulation and Evaluation Testing:** Validated a mask-aware expected cross-periodogram over 1,000 Monte Carlo simulations (MAD 0.00648) and analyzed missing-data effects on covariance parameter estimates.
- **Computational Implementation:** Implemented reusable Python/PyTorch pipelines with GLS mean profiling, L-BFGS optimization, CPU/GPU execution, and restartable HPC jobs.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)